

Teaching Guide — Week 1, Segment 1: Welcome & Roadmap (FULLY SHOWN)

This is the **one artifact shown at full depth** in the worked example. Everything else in the folder is a stub. Compare this to a stub file to see the difference between “sketched” and “done.”

Time: 0:00–0:20 (20 minutes) **Segment goal:** Set the tone, frame what the course is/isn’t, and state today’s objectives so students know exactly what success looks like.

Segment objectives

By the end of this segment, students will:

- Know the course is about **production-ready code**, not math derivation.
- Be able to state the five things they’ll be able to *do* by end of Week 1.
- Feel welcomed (whole-person tone set from minute one).

Materials

- Slide 1 (course roadmap) on the main monitor; this guide on the second monitor.
- Whiteboard for today’s objectives.

Verbatim script

(Say it in your own words — this is what to convey, not a script to read robotically.)

“Welcome to Introduction to Machine Learning Programming. Over the next 8 weeks we go from a simple straight line through data — today — all the way to neural networks.

Let me be clear about what this course **is** and **isn’t**. It **is** about building production-ready ML code — you’re domain experts building real proof-of-concepts. It is **not** a math course. You will *not* derive gradient descent. You **will** understand what it does well enough to use it and explain it.

By the end of *today*, you’ll be able to:

1. Run a complete ML pipeline — load, split, train, evaluate.
2. Train a linear regression model in sklearn.
3. Read residual plots to judge whether the model fits.
4. Explain a coefficient to your boss in plain English.

Quick show of hands — who’s used sklearn before? Who’s done any ML at all? Perfect, good mix. Let’s get started.”

Teaching tips

- **Energy sets the tone** — your enthusiasm in the first two minutes carries the room.
- Use names if you can (name tags / Zoom names) — whole-person care starts here.
- Point at the written objectives; refer back to them at the wrap-up.

Anticipated student questions (with answers)

- **“How much math do I really need?”** → “Intuition, not derivation. We focus on patterns and debugging.”
- **“Why sklearn and not PyTorch?”** → “sklearn first — simplest path to real pipelines. PyTorch/Keras come in Week 6.”

Timing contingencies

- **Running long?** Cut the show-of-hands; just state the objectives and move on.
- **Running short?** Take one extra question or briefly preview the 8-week arc.

Transition to Segment 2

“You now know where we’re headed today. Every ML project — yours included — follows the same pipeline. Let me draw it...” → go to Segment 2: The ML Pipeline.